



Sprint 2 Review: Modeling

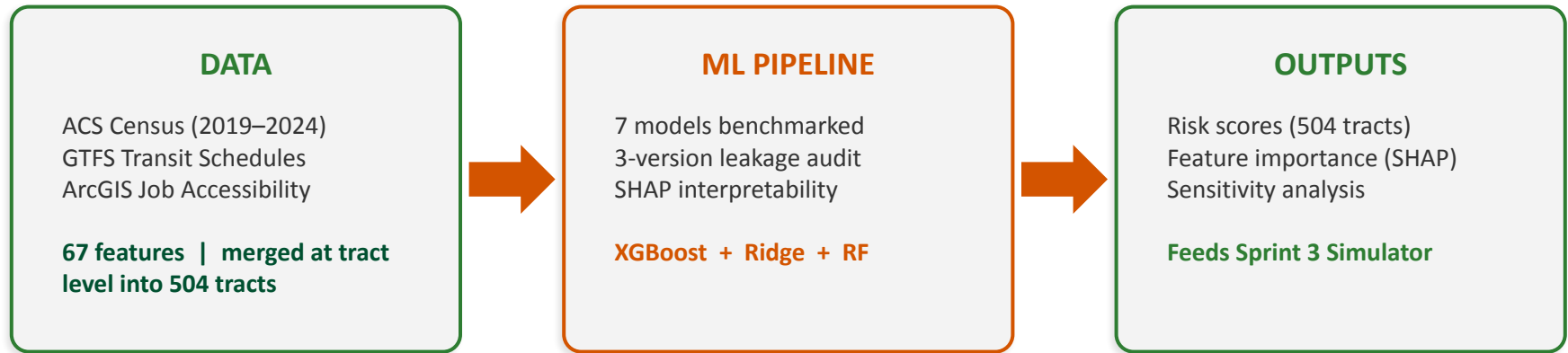
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Meeting Agenda

- 1 Problem Overview
- 2 Feature Consolidation
- 3 ACS Time Series
- 4 Regression and ML Modeling
- 5 Summary + Sprint 3

Predicting Transit Access Gaps

Across 504 Miami-Dade Census Tracts



Goal: Identify which GTFS levers (routes, stops, frequency) drive transit equity — so planners can prioritize interventions.

7

Equity Indicators

Transit Inequity = need x Access Deficit

504

Census Tracts

Miami-Dade County

67

Engineered Features

ACS + GTFS + ArcGIS

CLEAN MODEL

7 features | Zero leakage | Spatial neighbors + commute trends

EXTENDED MODEL

+6 GTFS features | Indirect leakage caveat | Headway, frequency, rail share

EXCLUDED

4 leaked features | Inside target formula | transit_jobs_30, stop_count, etc.

Why this matters

V2 model had $R^2=0.85$

but one leaked variable (transit_jobs_30) did 95% of the work.

V3 excludes all leaked features for honest, actionable predictions.

Demographic Trend Analysis

6-Year ACS Panel (2019–2024) | 707 Census Tracts | Linear Slope per Tract

Methodology

Variables tracked per tract:

Poverty rate, vehicle access, log income, senior share

Method:

scipy.stats.linregress on 6 annual observations per tract —
slope = annual rate of change

Forward Projection:

Extrapolate each tract's slope 3 years to 2027, anchored to
Sprint 2a equity tiers



Poverty Rate

-0.23 pp/yr

County median declining



No-Vehicle HH

±0.00 pp/yr

Flat median, rising in vulnerable tracts



Senior Share (65+)

+0.30 pp/yr

Growing aging population



Log Income

+0.06 /yr

Incomes slowly rising

County medians improve — but 161 tracts are worsening on both poverty AND vehicle access simultaneously.

Risk Scoring & Fragile Tracts

2027 Vulnerability Projections | Anchored to Sprint 2a Equity Tiers

22

Fragile Tracts

Moderate/High → Critical by 2027

161

Worsening Tracts

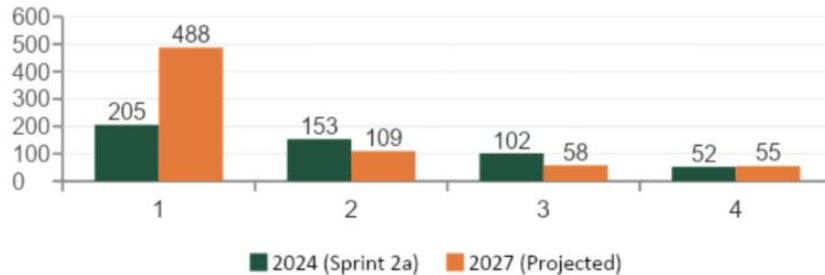
Poverty ▲ + No-Vehicle
▲ simultaneously

512

Tracts Scored

Equity tiers from Sprint 2a
baseline

Equity Tier Shift: 2024 → 2027



Fragile Tract Profile

	Fragile	Stable
Median Income	\$42.5K	\$85.5K
Uninsured Rate	19.6%	~10%
Unemployment	7.0%	4.2%
Rent Burden >30%	10.7%	8.9%

These 22 fragile tracts are the highest-ROI targets for Sprint 3 — intervention now can prevent Critical status by 2027.

Version 1

First regression model was circular and predicted the **equity _priority_score** using the same variables that built it

Artificially high **R^2 .93**

Focus: OLS, Ridge, Lasso, ElasticNet, Random Forest, Gradient Boosting



Version 2

Target the **composite_access_deficit** (the transit-only side) with only features a planner can actually observe or change

R^2 decreased to **.85**

Focus: OLS, Ridge, Lasso, ElasticNet, XGBoost



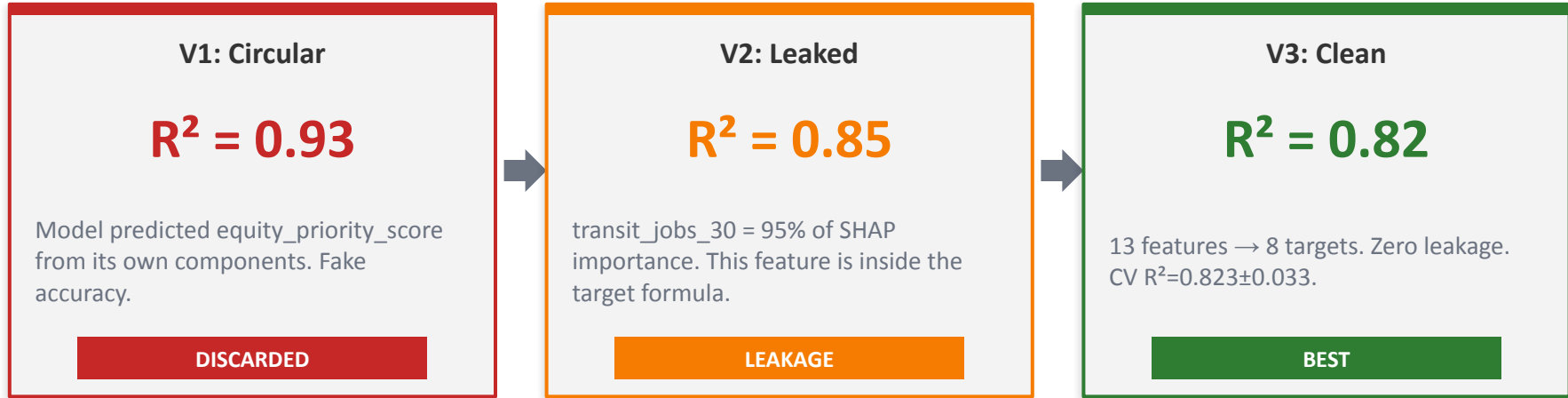
Version 3

$R^2 = 0.82$ (CV) | 0.79 (Test)

RMSE=0.056 |
MAE=0.045

Rebuilt 3/5 sub-indicators. Zero overlap.
13 features → 8 targets.
XGBoost + GridSearchCV.

Focus: Ridge, RandomForest, XGBoost



Key Findings

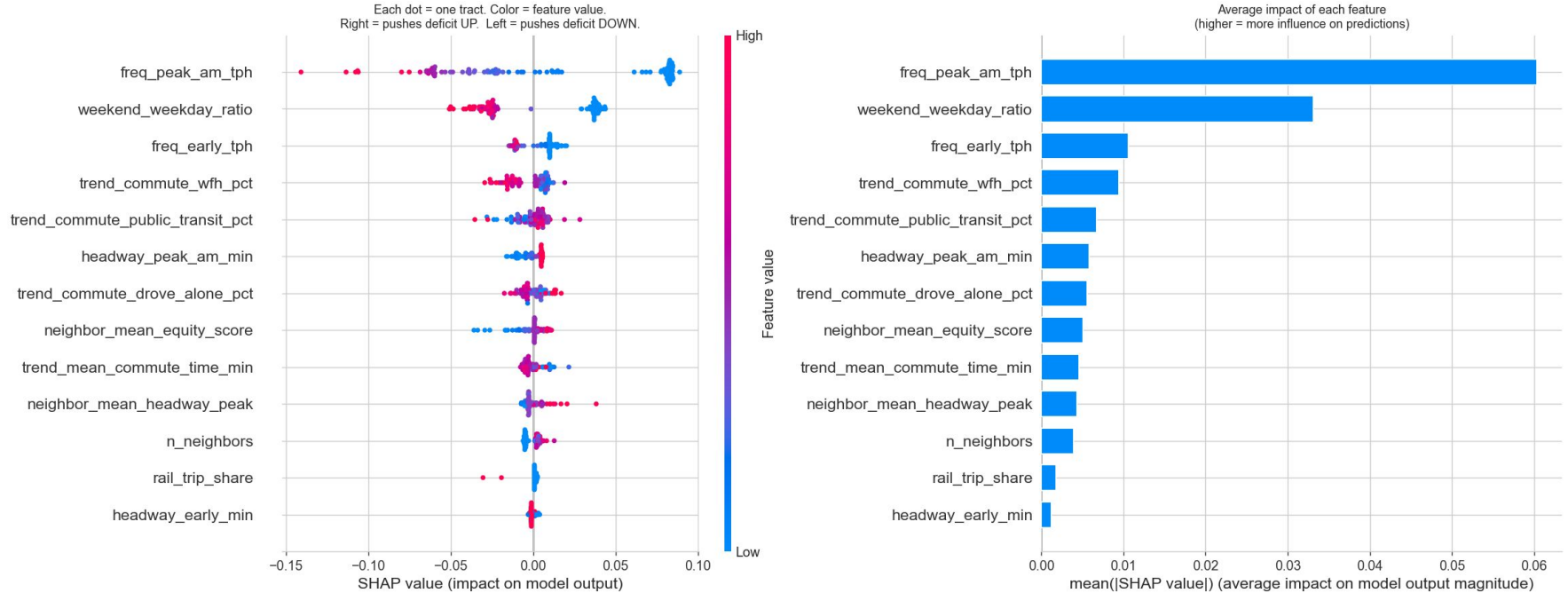
Model Performance: XGBoost best (CV $R^2=0.823$, Test $R^2=0.789$). Ridge=0.748, RF=0.806.

Top SHAP: AM peak freq (39.6%), weekend ratio (21.7%), early morning (6.9%)

Key Insight: 68% of predictions from just 2 features — AM peak freq & weekend ratio.

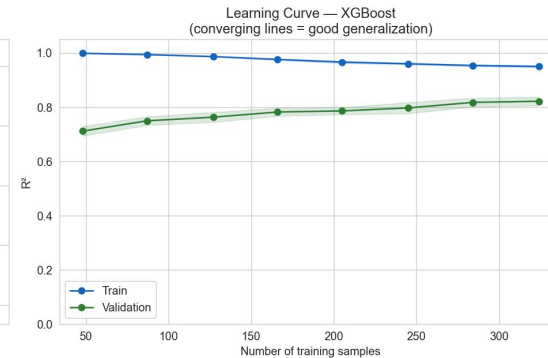
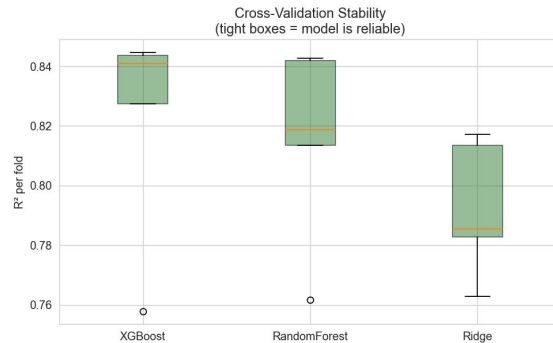
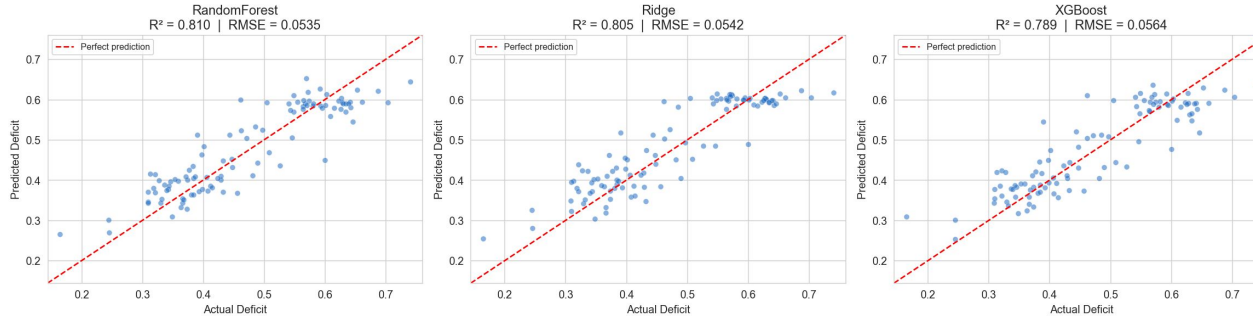
What-If: Top 3 combined (AM+25%, weekend parity, early+50%) → -0.078 deficit (14%).

SHAP Feature Importance



Predicted vs Actual & Learning Curves

How close are predictions to reality? (closer to red line = better)



Sprint 2 Delivered

- Leakage-free XGBoost model ($R^2=0.82$, RMSE=0.056)
- SHAP interpretatio
- Risk scores for all 504 census tracts
- What-if scenario analysis feeding Sprint 3 simulator
- ACS demographic trend forecasting

Sprint 3: What's Next

- Graph ML models (HuggingFace) for transit network
- What-If simulator + policy scenario testing
- Interactive equity dashboard (Streamlit/Gradio)
- GNN-based route optimization + access gap mapping
- Simulate service changes with trained models

S1: EDA

Jan–Feb

S2: Modeling

Feb–Mar

S3: Solution

Mar–Apr

S4: Dry-Run

Apr

S5: Final

May

Project Resources & Links

All deliverables, code, and documentation for the capstone project

GitHub Repository

55 issues (tasks) across 5 sprints

Notebooks, data, models

github.com/DanielRegaladoUMiami/CapstoneProject-AI-for-Equitable-Transportation

Data Sources

ACS Census • ArcGIS Accessibility

GTFS Transit • Miami-Dade MDT

[See Data/ folder in repository](#)

55 Issues • 5 Sprints • 8 Sub-sprints • 3 Decision Gates • Managed via GitHub Projects

Q&A